THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS TWO-CHAPTER TEST SET A

Chapters 1 & 2: **Patterns in Mathematics, Lines and Angles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The next number in the cube number sequence $1, 8, 27, 64, 125, \ldots$ is: 1

(a) 226 **(b)** 200 **(c)** 216 **(d)** 240

2. The next number in the powers of 3 sequence $1, 3, 9, 27, 81, \ldots$ is: 1

(a) 243 **(b)** 252 **(c)** 246 **(d)** 234

3. The sum of the first 6 odd numbers is: 1

(a) 36 **(b)** 49 **(c)** 21 **(d)** 30

4. Adding two consecutive triangular numbers 6 and 10 gives: 1

(a) 15 **(b)** 16 **(c)** 18 **(d)** 20

5. A complete graph with 4 points has how many lines joining them? 1

(a) 8 **(b)** 12 **(c)** 4 **(d)** 6

6. A line extends: 1

(a) endlessly in both directions **(b)** in one direction only

(c) in a closed curve **(d)** only between two points

7. Two lines that cross at a point are: 1

(a) rays **(b)** parallel lines

(c) intersecting lines **(d)** segments

8. The unit of angle measurement is: 1

(a) degree **(b)** centimetre **(c)** litre **(d)** gram

9. A straight angle measures: 1

(a) 180° **(b)** 90° **(c)** 270° **(d)** 360°

10. The angle between the hands of a clock at 6 o'clock is: 1

(a) 120° **(b)** 90° **(c)** 180° **(d)** 360°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State the rule of the sequence 2, 5, 8, 11, … and give the next term. 2

12. How many dots are there in the 6th triangular arrangement? 2

13. What are perpendicular lines? 2

14. Name the angle between the clock hands at 9 o'clock and its type. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Why does the powers-of-2 sequence grow so fast? Give the first 8 terms. 3

16. What is the sum of angles on a straight line? If one angle is 110°, find the other. 3

17. Write a rule of your own for a number pattern and give its first five terms. 3

OR

What is the connection between the number of sides of a regular polygon and the counting numbers? 3

18. What is a complete angle? How many right angles does it make? 3

OR

Arrange the angles in increasing order: 135°, 20°, 90°, 270°, 180°. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** Meena saves ₹1 on day 1, ₹2 on day 2, ₹4 on day 3, doubling each day. 5

(i) How much on day 6?
(ii) What sequence is this?
(iii) Total saved in 5 days?
(iv) On which day does she first save more than ₹100?

OR

(i) Explain with dots why the sum 1+2+3+2+1 equals 9. 5
(ii) Find 1+2+3+4+5+4+3+2+1.
(iii) Generalise for a sum up to 10 and down.

20. (i) State the sum of angles at a point. 5
(ii) If three angles at a point are 100°, 120° and x, find x.
(iii) Name the type of x.

OR

Case study: A student turns a rotating arm through 45°, 90°, 135° and 180° in turn. 5
(i) Name the types of the angles.
(ii) What fraction of a full turn is 135°?
(iii) How many steps of 45° complete a full turn?
(iv) Which angle is a right angle?

END OF QUESTION PAPER Total: 40 Marks

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